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Coupling energy-system models with a seam you can read and an agent you can audit

Active Inference in the NEST programme

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NEST is led by Open Earth Foundation with IIASA, CEU, Imperial College, C40, ICLEI, SCAC and the Active Inference Institute · Schmidt Sciences DEVI

ACTIVE INFERENCE IN THE NEST PROGRAMME · 6 MINUTES

A Vision for Resolving Nested Multi-Scale Modeling



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Decarbonisation decisions span nested scales that no single model covers

Decarbonization policies at local and global scales mutually affect one another

From household decisions to urban and national investment policies each require models that can transparently and rigorously capture the dynamics of their interdependencies.

National energy-system models set prices and fuel mixes.

City and building-stock models inform what gets built and retrofitted at one scale

Each carries its own assumptions, resolution and conventions.

A decision at one scale depends on results from the others.

Today model coupling depends upon manual processes and “expert” judgement

Hand-built links carry tacit conventions that are not written down.

Units, base years and geographies are manual and bespoke.

Two models can reach the same conclusion based upon different inputs.

Even model selection is not principled but part of the solver interface.

An agent in the loop raises the bar: every decision has to be inspectable

LLM orchestration can run the coupling faster than a person.

Yet they are black boxes that make assumptions and decisions nobody monitored.

An outcome you cannot trace, audit or verify is a result you cannot trust or act on.

Policy makers rely upon models they can trust whose results are replicable, transparent, auditable and verifiable.

As a first step NEST makes model boundaries explicit, and then tests them

What is Declared

The boundary between models: every exchanged variable carries its unit, convention, geography and mapping rule, sourced to its owner.

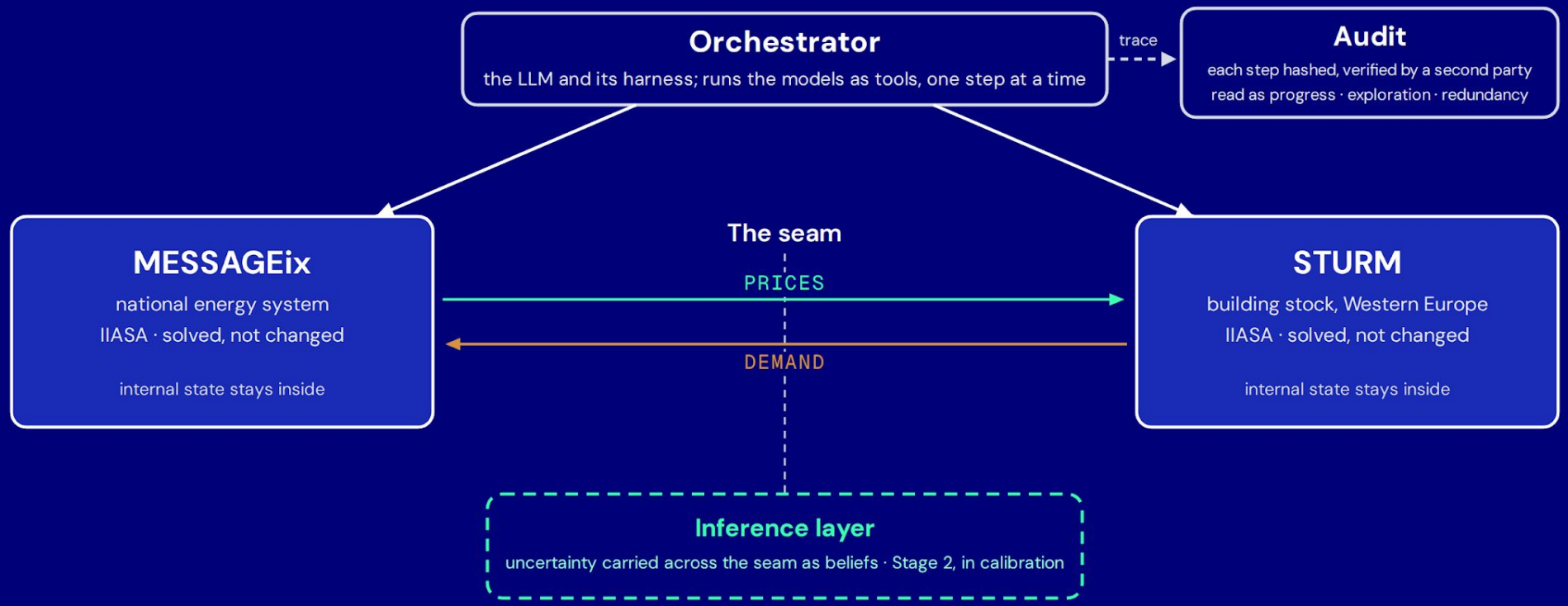
The uncertainty is made explicit and tested: a belief layer specified once in GNN, calibrated before it is trusted.

What is Audited

The agent: a trace contract that an independent reader checks for schema, hashes and accounting.

The first runs: two real exports pass those checks. They do not yet validate the energy computation.

NEST in one picture: the models stay whole, an agent runs them, every step is audited



Both IIASA models run as tools, unchanged. Prices cross one way and demand the other. The agent's steps are hashed and read by a second party.

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The vision: a federation of project agents across the DEVI cohort

Provide a principled architecture for nested models with explicit boundaries.

Couplings leave inspectable records.

Different groups can test and reason about different local and global models as a coherent whole.

Active Inference provides a coherent and principled architecture for the federation, alignment and synchronisation of different models at different scales

What we do not claim yet

No validation against observed energy or building data.

No speed-up over the hand-built coupling.

The first result is expected to match the hand-tuned loop.

Whether an agent can work out a coupling from the goal alone is the next experiment.